



PT. Century Batteries Indonesia

# Quality Mapping Area : Charging A



## I. Acid Filling

| Acid Filling - Wet A                                                              |                     |                             |                                   |                   |               |     |          |
|-----------------------------------------------------------------------------------|---------------------|-----------------------------|-----------------------------------|-------------------|---------------|-----|----------|
| Drawing / Sketch                                                                  | Rank                | Quality Requirement         |                                   | Control Instrumen |               |     |          |
|                                                                                   |                     | Item Check                  | Standard                          | Gauge/Tools       | Frequency     | PIC | Pokayoke |
| MF NS40 Series                                                                    |                     |                             |                                   |                   |               |     |          |
|  | Item Check - Produk |                             |                                   |                   |               |     |          |
|                                                                                   | 1                   | Ketinggian Level Elektrolit | Garis UL (33 - 35 dari separator) | Jig               | 1X/2jam       | P   |          |
|                                                                                   |                     |                             |                                   |                   | 2X/Shift/type | Q   |          |
|                                                                                   | 2                   | SpGr Battery                | 1,215 - 1,225                     | Hidrometer        | 1X/2jam       | P   |          |
|                                                                                   |                     |                             |                                   |                   | 2X/Shift/type | Q   |          |
|                                                                                   | Item Check - Proses |                             |                                   |                   |               |     |          |
|                                                                                   | 1                   | Waktu timer filling         | 5 - 15 sekon                      | Display setting   | 1X/2jam       | P   |          |
|                                                                                   | 2                   | SpGr bak Mesin              | 1,215 - 1,225                     | Hidrometer        | 1X/2jam       | P   |          |
|                                                                                   |                     |                             |                                   |                   | 2X/Shift/type | Q   |          |



## II. Charging

| Charging - Wet A                                                                    |      |                     |                        |                                                                                                                                                |                             |                                                      |                          |  |
|-------------------------------------------------------------------------------------|------|---------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------|--------------------------|--|
| Drawing / Sketch                                                                    | Rank | Quality Requirement |                        | Control Instrumen                                                                                                                              |                             |                                                      |                          |  |
|                                                                                     |      | Item Check          | Standard               | Gauge/Tools                                                                                                                                    | Frequency                   | PIC                                                  | Pokayo                   |  |
| MF NS40 Series                                                                      |      |                     |                        |                                                                                                                                                |                             |                                                      |                          |  |
|  |      |                     |                        |                                                                                                                                                |                             |                                                      |                          |  |
| Item Check - Produk                                                                 |      |                     |                        |                                                                                                                                                |                             |                                                      |                          |  |
|                                                                                     |      | 1                   | SpGr                   | 1,215 - 1,225                                                                                                                                  | Hidrometer                  | 1X/2jam<br>2X/Shift/type                             | (P)<br>(Q)               |  |
| Item Check - Proses                                                                 |      |                     |                        |                                                                                                                                                |                             |                                                      |                          |  |
|                                                                                     |      | 1                   | Parameter Charging     | Arus Charging dari program database<br>Temperature Cooling : max 35 C<br>Temperature Battery : max 60 C<br>Temperature saat finishing : max 35 | Tang Amperes<br>Thermometer | 1X/2jam<br>2X/Shift/type<br>1X/2jam<br>2X/Shift/type | (P)<br>(Q)<br>(P)<br>(Q) |  |
|                                                                                     |      | 3                   | Susunan Jumlah Battery | Type Battery : 1800 Series<br>Jumlah : 1800 pcs                                                                                                | Visual                      | 100%                                                 | (P)                      |  |



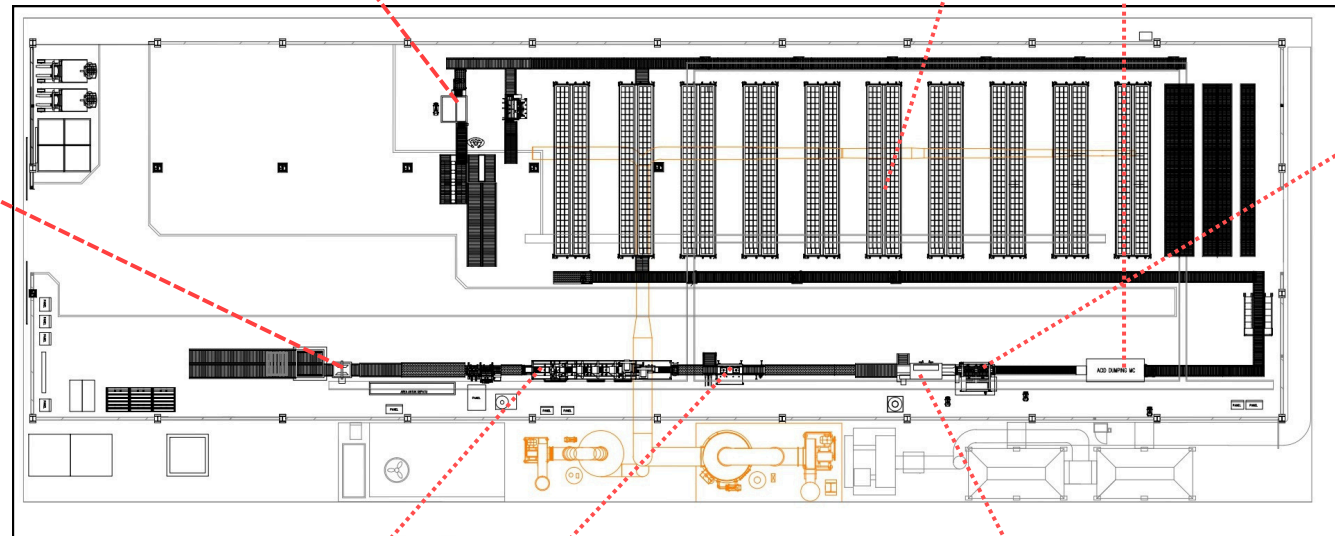
## III. Dumping

| Dumping - Wet A                                                                     |                     |                      |                |                                      |                       |         |          |  |
|-------------------------------------------------------------------------------------|---------------------|----------------------|----------------|--------------------------------------|-----------------------|---------|----------|--|
| Drawing / Sketch                                                                    | Rank                | Quality Requirement  |                | Control Instrumen                    |                       |         |          |  |
|                                                                                     |                     | Item Check           | Standard       | Gauge/Tools                          | Frequency             | PIC     | Pokayoke |  |
| MF NS40 Series                                                                      |                     |                      |                |                                      |                       |         |          |  |
|  | Item Check - Produk |                      |                |                                      |                       |         |          |  |
|                                                                                     |                     | 1                    | Volume Asam    | Tidak ada Asam dalam battery         | Visual                | 1x/2jam | (P)      |  |
|                                                                                     | Item Check - Proses |                      |                |                                      |                       |         |          |  |
|                                                                                     |                     | 1                    | Jumlah Battery | Parameter Into No. 4 - 6 pcs / cycle | Visual                | 1X/2jam | (P)      |  |
|                                                                                     | 2                   | waktu proses dumping |                | Parameter Empty T. 5 - 7 s           | Display setting mesin | 1X/2jam | (P)      |  |
|                                                                                     |                     |                      |                | Into Delay 5 - 7 s                   |                       |         |          |  |
|                                                                                     |                     |                      |                | Acid Dump 5 - 7 s                    |                       |         |          |  |
|                                                                                     |                     |                      |                |                                      |                       |         |          |  |



## IV. Acid Levelling

| PLT & Packaging - Wet A                                                            |                     |                         |                                           |                                                                                                                                                                                                                                                                                                                   |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|------------------------------------------------------------------------------------|---------------------|-------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------|----------|------|-----|----|---------------|-----------|----------|---------------|--------------------------------------------|--|--|--|--|--------|---------|---|--|
| Drawing / Sketch                                                                   | Rank                | Quality Requirement     |                                           | Control Instrumen                                                                                                                                                                                                                                                                                                 |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|                                                                                    |                     | Item Check              | Standard                                  | Gauge/Tools                                                                                                                                                                                                                                                                                                       | Frequency    | PIC                   | Pokayoke |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
| MF NS40 Series                                                                     |                     |                         |                                           |                                                                                                                                                                                                                                                                                                                   |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|  | Item Check - Produk |                         |                                           |                                                                                                                                                                                                                                                                                                                   |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|                                                                                    |                     | 1                       | Packaging                                 | Appearance Battery                                                                                                                                                                                                                                                                                                | Visual       | 100%<br>1X/shift/type | Ⓟ        |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|                                                                                    | Item Check - Proses |                         |                                           |                                                                                                                                                                                                                                                                                                                   |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|                                                                                    |                     | 1                       | Parameter PLT                             | <table border="1"><thead><tr><th>Type Battery</th><th>OV</th><th>CV2</th><th>▲ CV</th><th>CV3</th></tr></thead><tbody><tr><td>MF</td><td>12.75 - 13.00</td><td>8.0 - 8.8</td><td>Max 0.15</td><td>12.30 - 13.00</td></tr><tr><td colspan="5">Standar Maksimal OV dan CV3 adalah 13 volt</td></tr></tbody></table> | Type Battery | OV                    | CV2      | ▲ CV | CV3 | MF | 12.75 - 13.00 | 8.0 - 8.8 | Max 0.15 | 12.30 - 13.00 | Standar Maksimal OV dan CV3 adalah 13 volt |  |  |  |  | Visual | 1x/2jam | Ⓟ |  |
|                                                                                    | Type Battery        | OV                      | CV2                                       | ▲ CV                                                                                                                                                                                                                                                                                                              | CV3          |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
| MF                                                                                 | 12.75 - 13.00       | 8.0 - 8.8               | Max 0.15                                  | 12.30 - 13.00                                                                                                                                                                                                                                                                                                     |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
| Standar Maksimal OV dan CV3 adalah 13 volt                                         |                     |                         |                                           |                                                                                                                                                                                                                                                                                                                   |              |                       |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |
|                                                                                    | 2                   | Verifikasi alat deteksi | Buzzer / alarm berbunyi ketika NG process | Visual                                                                                                                                                                                                                                                                                                            | 1X/2jam      | Ⓟ                     |          |      |     |    |               |           |          |               |                                            |  |  |  |  |        |         |   |  |



## Acid Levelling - Wet A

Drawing / Sketch

Rank

Quality Requirement

Item Check

Control Instrumen

Gauge/Tools

Frequency

PIC

Pokayoke

MF NS40 Series

### Item Check - Produk

1 Kekencangan Vent Plug

Tightening Torque

5-9 kgf/cm  
(Daihatsu-Isuzu)

Torque Meter

1X/2jam

P

2 SpGr Battery

Proses

Standar SpGr

One shot & Battery OEM

Two Shot

1,275 ± 0,005

1,285 ± 0,005

Hidrometer

1X/2jam

P

Q

### Item Check - Proses

1 Waktu Levelling

Parameter

Filling First

Vacuum First

Filling 2nd

Vacuum 2nd

Nilai Setting

Type Small

85-125

-

-

35-55

Display Setting Mesin

1X/2jam

P

2 Screw Ventplug

Parameter

Nilai Setting

Tekanan Angin

Proses

Standar SpGr

0.3 - 0.5 Mpa

Regulator Pressure

1X/2jam

P

3 SpGr Bak

One shot & Battery OEM

Two Shot

1,310 ± 0,005

1,310 ± 0,005

Hidrometer

1X/2jam

P



## VII. Washing & Drying

| Washing & Drying- Wet A                                                            |                    |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|------------------------------------------------------------------------------------|--------------------|---------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------|---------|----------------|-----------|-----------|---------|------------|----------|----------------|----------|--------|---------|---|--|
| Drawing / Sketch                                                                   | Rank               | Quality Requirement |                         | Control Instrumen                                                                                                                                                                                                                                                                                        |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|                                                                                    |                    | Item Check          | Standard                | Gauge/Tools                                                                                                                                                                                                                                                                                              | Frequency | PIC           | Pokayoke    |         |                |           |           |         |            |          |                |          |        |         |   |  |
| MF NS40 Series                                                                     |                    |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|  | Item Check- Produk |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|                                                                                    |                    | 1                   | Appreance Battery       | Tidak basah dan Bersih                                                                                                                                                                                                                                                                                   | Visual    | 100%          | P           |         |                |           |           |         |            |          |                |          |        |         |   |  |
|                                                                                    | Item Check- Proses |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|                                                                                    |                    | 1                   | Parameter Timer Washing | <table><tr><th>Parameter</th><th>Nilai Setting</th></tr><tr><td>Delay Input</td><td>5 detik</td></tr><tr><td>LR Brush Delay</td><td>1.5 detik</td></tr><tr><td>Huff Time</td><td>5 Detik</td></tr><tr><td>Over Delay</td><td>10 detik</td></tr><tr><td>Fan Stop Delay</td><td>10 detik</td></tr></table> | Parameter | Nilai Setting | Delay Input | 5 detik | LR Brush Delay | 1.5 detik | Huff Time | 5 Detik | Over Delay | 10 detik | Fan Stop Delay | 10 detik | Visual | 1X/2jam | P |  |
|                                                                                    | Parameter          | Nilai Setting       |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
| Delay Input                                                                        | 5 detik            |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
| LR Brush Delay                                                                     | 1.5 detik          |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
| Huff Time                                                                          | 5 Detik            |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
| Over Delay                                                                         | 10 detik           |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
| Fan Stop Delay                                                                     | 10 detik           |                     |                         |                                                                                                                                                                                                                                                                                                          |           |               |             |         |                |           |           |         |            |          |                |          |        |         |   |  |
|                                                                                    | 2                  | Suhu Drying         | Range Suhu 60 - 70 C    | Visual                                                                                                                                                                                                                                                                                                   | 1X/2jam   | P             |             |         |                |           |           |         |            |          |                |          |        |         |   |  |



## VI. Air Leak Test

| Air Leak Test (ALT) - Wet A                                                           |                     |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|---------------------------------------------------------------------------------------|---------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|----------------------|----------|-----------------------|---------|-----------------|---------|--------------|----|-----------|----|--------|---------|---|--|
| Drawing / Sketch                                                                      | Rank                | Quality Requirement     |                                                                                                                                                                                                                                                                                                              | Control Instrumen |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|                                                                                       |                     | Item Check              | Standard                                                                                                                                                                                                                                                                                                     | Gauge/Tools       | Frequency     | PIC                  | Pokayoke |                       |         |                 |         |              |    |           |    |        |         |   |  |
| MF NS40 Series                                                                        |                     |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|  | Item Check - Produk |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|                                                                                       | 1                   | Appreance Battery       | tidak keluar asam dari celah container cover ataupun dari lubang gas vent                                                                                                                                                                                                                                    | Visual            | 2x/shift      | P                    |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|                                                                                       | Item Check - Proses |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|                                                                                       | 1                   | Verifikasi alat deteksi | Buzzer / alarm berbunyi ketika NG process                                                                                                                                                                                                                                                                    | Dummy Battery     | 1X/shift/type | Q                    |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
|                                                                                       | 2                   | Parameter ALT           | <table><tr><th>Parameter</th><th>Nilai Setting</th></tr><tr><td>Pressure upper limit</td><td>20 kPa</td></tr><tr><td>Differential pressure</td><td>0.3 kPa</td></tr><tr><td>Air-filled time</td><td>5 - 8 s</td></tr><tr><td>Balance time</td><td>1s</td></tr><tr><td>Test time</td><td>3s</td></tr></table> | Parameter         | Nilai Setting | Pressure upper limit | 20 kPa   | Differential pressure | 0.3 kPa | Air-filled time | 5 - 8 s | Balance time | 1s | Test time | 3s | Visual | 1X/2jam | P |  |
| Parameter                                                                             | Nilai Setting       |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
| Pressure upper limit                                                                  | 20 kPa              |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
| Differential pressure                                                                 | 0.3 kPa             |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
| Air-filled time                                                                       | 5 - 8 s             |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
| Balance time                                                                          | 1s                  |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |
| Test time                                                                             | 3s                  |                         |                                                                                                                                                                                                                                                                                                              |                   |               |                      |          |                       |         |                 |         |              |    |           |    |        |         |   |  |



## V. Level Checker

| Level Checker - Wet A                                                                 |                     |                         |                                           |                            |                |                       |               |                       |         |
|---------------------------------------------------------------------------------------|---------------------|-------------------------|-------------------------------------------|----------------------------|----------------|-----------------------|---------------|-----------------------|---------|
| Drawing / Sketch                                                                      | Rank                | Quality Requirement     |                                           | Control Instrumen          |                |                       |               |                       |         |
|                                                                                       |                     | Item Check              | Standard                                  | Gauge/Tools                | Frequency      | PIC                   | Pokayoke      |                       |         |
| MF NS40 Series                                                                        |                     |                         |                                           |                            |                |                       |               |                       |         |
|  | Item Check - Produk |                         |                                           |                            |                |                       |               |                       |         |
|                                                                                       | 1                   | Level Elektrolit        | Type                                      | Standar level              | Jig Visual     | 100%<br>2x/shift/type | P             |                       |         |
|                                                                                       |                     |                         | NS40                                      | 34± 1 mm di atas separator |                |                       | Q             |                       |         |
|                                                                                       | Item Check - Proses |                         |                                           |                            |                |                       |               |                       |         |
|                                                                                       | 1                   | Verifikasi alat deteksi | Buzzer / alarm berbunyi ketika NG process |                            |                | Dummy Battery         | 2X/shift/type | Q                     |         |
|                                                                                       |                     |                         | Type                                      | Lebar Conveyor             | Input Conveyor |                       |               | Display Setting Mesin | 1X/2jam |
|                                                                                       | 2                   | Parameter mesin         | NS40                                      | 6.5                        | 13.5           |                       |               |                       |         |